

Python Programming

Overview

Python is a high-level, general-purpose language known for readable syntax and a large ecosystem. It is widely used in data science, machine learning, web development, automation, scientific computing, and scripting.

Variables and Data Structures

Python uses dynamic typing, so variables do not need explicit type declarations. Important built-in structures include lists, dictionaries, tuples, and sets. Dictionaries store key-value pairs and are especially useful for fast lookups.

Functions

Functions organize reusable code. They can accept parameters and return values. Python also supports lambda expressions, default parameters, keyword arguments, and higher-order functions.

Object-Oriented Programming

Python supports classes and objects. Classes can contain attributes and methods and can use inheritance to build relationships between types.

Applications

Python is particularly popular in artificial intelligence and data science. Libraries such as NumPy, pandas, scikit-learn, and PyTorch provide tools for numerical computing, data analysis, machine learning, and deep learning.